PHY 1110
Experiment 4: Projectile Motion

Purpose:

The purpose of this experiment is to study the motion of a projectile and to predict its path.

Your grade for this experiment will be determined by the success of your prediction.

Equipment:

Spring Loaded Projectile Launcher, Catch Box, carbon copy paper, regular paper and Meter Sticks.

Introduction:

In this experiment, an object will be launched by a spring gun. Once it is released, the object moves under the influence of gravity alone, if the effects of air resistance are neglected. We will assume that air resistance is small enough to be neglected. In a later experiment, you will study the effects of air resistance.

Because the force of gravity is exactly vertical, under the influence of gravity alone, an object will experience a constant vertical acceleration and no horizontal acceleration. Using a coordinate system with a vertical y-axis and a horizontal x-axis, $a_y = -g$ and $a_x = 0$, the equations of motion for a projectile are then:

$$v_x = v_{xo} = constant \tag{1}$$

$$v_y = v_{yo} - gt \tag{2}$$

and

$$x = x_o + v_{xo}t \tag{3}$$

$$y = y_o + v_{yo}t - \frac{1}{2}gt^2 \tag{4}$$

As can be seen from Eq. (3) and (4), the path of the projectile, $x(t)$ and $y(t)$ can always be calculated if we know the initial position [x_o, y_o] and the components of the initial velocity [v_{xo},v_{yo}].

The experiment:

The projectile will be fired by a spring gun. Your assignment is to predict the height of the projectile after it has traveled a fixed horizontal distance specified by your instructor, typically 150 cm.

NOTE: The spring gun you will use is part of the "Ballistic Pendulum Apparatus". Do not remove the pendulum from the stand. It will be used for other experiments.

If you prop up the gun as shown in Figure 1 below, you can easily determine the initial position of the projectile. But, you cannot directly determine the components of the initial velocity.

If the angle θ at which the projectile is launched is known, the velocity components can be determined from the magnitude of the initial velocity (the speed), v_o.

$$v_{xo} = v_o \cos(\theta) \tag{5}$$

$$v_{yo} = v_o \sin(\theta) \tag{6}$$

So, first you must determine v_o !!

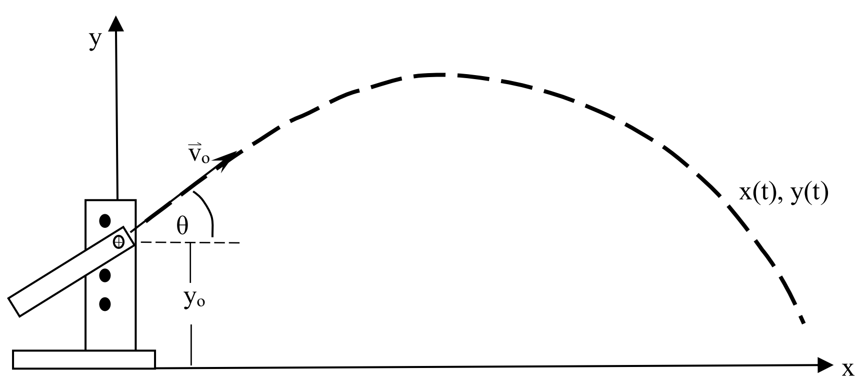

Figure 1: Projectile launched at an angle θ above the surface of a table.

Finding the Initial Speed of the Projectile

The initial speed can most easily be found by observing how far the projectile travels when launched horizontally from a known height. See Figure 2 below.

Launching the projectile horizontally gives $v_{xo} = v_o$ and $v_{yo} = 0$. In this case, Equations (3) and (4) simplify to become

$$x = x_o + v_o t \tag{7}$$

$$y = y_o - \frac{1}{2} g t^2 \tag{8}$$

Launch the projectile into the catch box several times until you get a set of consistent range values for your projectile.

Obtain the average value of the range, $\bar{x}$, and the uncertainty of the range, $u_{\bar{x}}$.

Use Equations (7) and (8) to calculate the time that your projectile spends in the air. Then obtain the average initial speed $\bar{v}_o$, and the uncertainty of the initial speed $u_{\bar{v}_o}$.

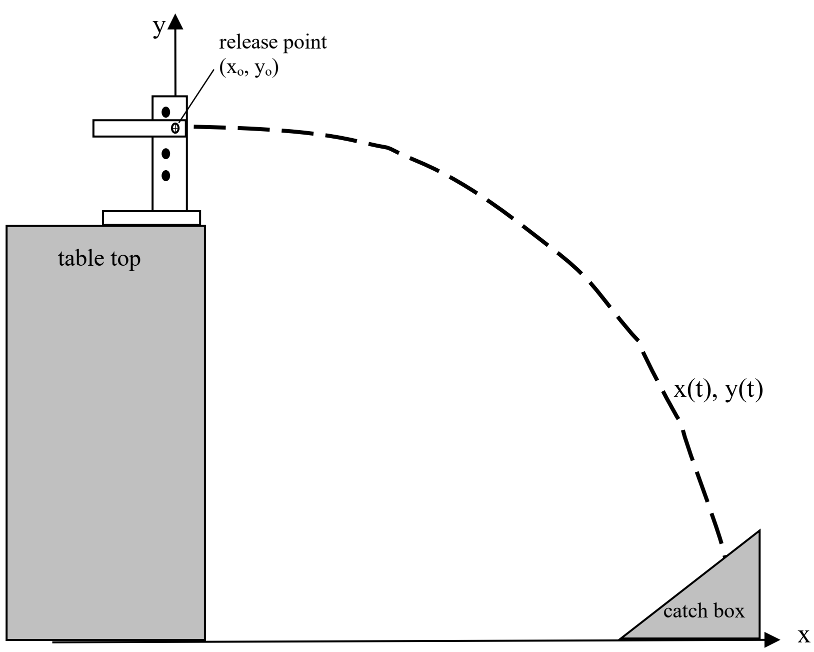

Figure 2: The experimental arrangement for determining the initial velocity of the projectile.

Predicting the height of the projectile

Once you have determined the initial speed of the projectile, set up the spring launcher as illustrated in Figure 1.

Prop up the gun platform at one end of the table, making sure it is at a stable angle. Direct the projectile toward the other end of the table. By appropriate measurements and a little bit of trigonometry, determine the angle of inclination of the gun, θ, as well as the initial launch height of the projectile above the table. Determine the x- and y-components of the initial velocity. This velocity has the same magnitude as before, but now you must use the angle θ and trigonometry to determine its components v_{xo} and v_{yo}.

Use Equations (3) and (4) to calculate the height of the projectile **above the table**, at a horizontal distance *x* of 150 cm from the launch point (unless your instructor tells you otherwise). This may or may not be y, depending upon your choice of the location for *y* = 0. Be careful of mixed units: don't mix centimeters and meters!

When you are confident of your prediction, notify the instructor. The instructor will elevate a ring-stand to the distance **above the table** that you specify.

You will have to estimate the correct azimuth (angle in the horizontal plane) so that the projectile will remain in the correct vertical plane. If you have done your calculations correctly, the projectile will easily pass through the center of the ring!

The accuracy of your prediction will determine the grade for your laboratory group.